Asta

Immature Neuron Cell Types and Development Across Brain Regions

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Introduction and Background

The brain has several regions where new neurons continue to form and mature throughout life, with different areas producing either excitatory or inhibitory cell types. These immature neurons go through specific developmental stages and have distinct names based on their maturation level and brain region. (LLM Memory)

Adult neurogenesis, the formation of new neurons in the mature brain, occurs in specific brain regions where immature neurons can be found at various developmental stages (Model-Generated). These immature neurons are classified based on their developmental stage, morphological characteristics, and functional properties (Model-Generated). The classical nomenclature includes neuroblasts (the earliest migrating cells), immature neurons (cells that have begun to differentiate but lack full maturation), and region-specific subtypes based on their final cell identity (Model-Generated). In excitatory regions like the dentate gyrus and piriform cortex, immature neurons develop into glutamatergic granule cells and pyramidal neurons respectively (Model-Generated). In contrast, regions like the olfactory bulb, striatum, and cerebral cortex receive immature inhibitory interneurons that differentiate into various GABAergic subtypes (Model-Generated). The developmental progression typically involves migration from neurogenic niches, followed by morphological maturation, synapse formation, and functional integration into existing circuits (Model-Generated). Understanding these cell types and their developmental stages is crucial for comprehending how the adult brain maintains plasticity and repairs neural circuits (Model-Generated).

Dentate Gyrus Immature Neurons

The dentate gyrus produces new excitatory granule cells through a well-defined developmental sequence starting with neural stem cells in the subgranular zone that progress through intermediate progenitors, neuroblasts, and immature granule cells before becoming mature neurons. These cells migrate only short distances and follow a specific transcription factor cascade during development. (10 sources)

In the dentate gyrus, adult neurogenesis begins with neural stem cells in the subgranular zone (SGZ) that give rise to excitatory granule neurons through a highly organized developmental progression (Velusamy et al., 2017) (Regalado-Santiago et al., 2016). The classical nomenclature identifies several distinct cell types based on their developmental stage and molecular markers. Type I cells are radial glial-like stem cells expressing GFAP and Sox2, while type II cells are non-sessile transiently activated progenitor cells expressing nestin (Stepien, 2021) (Attardo et al., 2009).

The developmental sequence progresses through well-defined stages with specific molecular signatures. Neural stem cells develop into proliferating neural progenitor cells, including type-2a cells (Nestin+/Sox2+), type-2b cells (Nestin+/DCX+), and type-3 neuroblasts (DCX+) (Micheli et al., 2025). Type-3 progenitor cells then differentiate into post-mitotic immature granule cells and finally into mature granule cells that populate the granule cell layer (Tanaka et al., 2019) (Hodge et al., 2008). Neuroblasts progress toward immature postmitotic granule neurons co-expressing DCX and NeuN (stage 5), and eventually become terminally differentiated neurons (stage 6) expressing calbindin and NeuN (Micheli et al., 2025).

Unlike other neurogenic regions, dentate gyrus cells migrate only a few micrometers from the subgranular zone to the granule cell layer (Vik-Mo et al., 2012). The immature neurons migrate radially outward as they differentiate and eventually receive primary glutamatergic afferents from entorhinal cortex while projecting axons to inhibitory interneurons and pyramidal cells of area CA3 (Stoll, 2014) (Zhao et al., 2006). A notable transcription factor cascade (Pax6 → Ngn2 → Tbr2 → NeuroD → Tbr1) controls neurogenesis in the adult SGZ, similar to embryonic cerebral cortex development (Hodge et al., 2008).

Evidence

(Velusamy et al., 2017)

Protective Effect of Antioxidants on Neuronal Dysfunction and Plasticity in Huntington's Disease

> *"In the hippocampal SGZ, proliferating NSCs develop into intermediate progenitors, which generate neuroblasts or immature neurons. These newly generated immature neurons migrate into the inner granule cell layer (GCL) and differentiate into new granule neurons of the hippocampus (Ming et al., 2005)"*

> *"In the SVZ, proliferating radial glia-like cells give rise to transient amplifying cells that generate migrating neuroblasts (Ming et al., 2005)(Lois et al., 1996). Through the rostral migratory system (RMS), neuroblasts migrate towards the olfactory bulb (OB) (Lois et al., 1996). In the OB, immature neurons differentiate into different subtypes of interneurons in granule cell layer (GCL) and glomerular layer (GLOM)."*

(Regalado-Santiago et al., 2016)

Mimicking Neural Stem Cell Niche by Biocompatible Substrates

> *"In the rostral migratory stream (RMS), type A cells form a chain and migrate toward the olfactory bulb through a tube formed by astrocytes. Upon reaching the olfactory bulb, immature neurons leave the RMS and migrate radially toward the glomeruli, where they differentiate into different subtypes of interneurons that, finally, are synaptically integrated"*

> *"hippocampal neurogenesis in the DG begins, as in SVZ neurogenesis, with the proliferation of radial (type I cell) and nonradial (type II cell) precursors that give rise to intermediate progenitors, which in turn generate neuroblasts"*

> *"The new immature neurons move into the inner granule cell layer and differentiate into dentate granule cells in the hippocampus."*

(Stepien, 2021)

Neurogenesis in neurodegenerative diseases in the adult human brain

> *"Three types of proliferatively active cells have been identified in the granular layer of the dentate gyrus (DG) of the hippocampus: type I cells -radial glial-like stem cells expressing glial fibrillary acidic protein (GFAP) and Sox2; type II cells -non-sessile cells expressing nestin, also referred to as transiently activated progenitor cells, neuroblasts expressing doublecortin (DCX); and Ki67 proteins and immature neurons expressing the DCX protein, PSA-NCAM, a marker of migrating neurons (polysialylated neuronal cell adhesion molecules) and neuron-specific protein (NeuN) (Attardo et al., 2009)(Gault et al., 2021). On the other hand, three types of cells were distinguished in the subventricular zone (SVZ): B1-type astrocytic stem cells, GFAP-positive, C-type progenitor cells expressing the Mash1 protein, and neuroblasts expressing the DCX protein (Okano et al., 2008)."*

(Attardo et al., 2009)

Tis21 Expression Marks Not Only Populations of Neurogenic Precursor Cells but Also New Postmitotic Neurons in Adult Hippocampal Neurogenesis

> *"During embryonic cortical development, expression of Tis21 is associated with cell cycle lengthening and neurogenic divisions of progenitor cells. We here investigated if the expression pattern of Tis21 also correlates with the generation of new neurons in the adult hippocampus. We used Tis21 knock-in mice expressing green fluorescent protein (GFP) and studied Tis21-GFP expression together with markers of adult hippocampal neurogenesis in newly generated cells. We found that Tis21-GFP 1) was absent from the radial glia–like putative stem cells (type-1 cells), 2) first appeared in transient amplifying progenitor cells (type-2 and 3 cells), 3)*

did not colocalize with markers of early postmitotic maturation stage, 4) was expressed again in maturing neurons, and 5) finally decreased in mature granule cells. Our data show that, in the course of adult neurogenesis, Tis21 is expressed in a phase additional to the one of the embryonic neurogenesis. This additional phase of expression might be associated with a new and different function of Tis21 than during embryonic brain development, where no Tis21 is expressed in mature neurons. We hypothesize that this function is related to the final functional integration of the newborn neurons. Tis21 can thus serve as new marker for key stages of adult neurogenesis."

(Micheli et al., 2025)

Survey of transcriptome analyses of hippocampal neurogenesis with focus on adult dentate gyrus stem cells

"Neural stem cells (NSCs) progressively develop into proliferating neural progenitor cells (NPCs), designated as type-2a (Nestin + / Sox2 +), type-2b cells (expressing Nestin and doublecortin: Nestin + /DCX +) and neuroblasts (type-3, DCX +) (Filippov et al., 2003) (Fukuda et al., 2003)(Kronenberg et al., 2003)(Steiner et al., 2006). Neuroblasts progress toward immature postmitotic granule neurons co-expressing DCX and NeuN (stage 5), and eventually become terminally differentiated neurons (stage 6) expressing calbindin and NeuN (Brandt et al., 2003)Steiner et al., 2004)."

(Tanaka et al., 2019)

Ameliorating effect of postweaning exposure to antioxidant on disruption of hippocampal neurogenesis induced by developmental hypothyroidism in rats.

"Type-3 progenitor cells differentiate into post-mitotic immature granule cells and finally into mature granule cells that populate the granule cell layer (GCL) (Hodge et al., 2008;Sibbe and Kulik, 2017)."

(Hodge et al., 2008)

Intermediate Progenitors in Adult Hippocampal Neurogenesis: Tbr2 Expression and Coordinate Regulation of Neuronal Output

"Neurogenesis in the adult hippocampus is a highly regulated process that originates from multipotent progenitors in the subgranular zone (SGZ). Currently, little is known about molecular mechanisms that regulate proliferation and differentiation in the SGZ. To study the role of transcription factors (TFs), we focused on Tbr2 (T-box brain gene 2), which has been implicated previously in developmental glutamatergic neurogenesis. In adult mouse hippocampus, Tbr2 protein and Tbr2-GFP (green fluorescent protein) transgene expression were specifically localized to intermediate-stage progenitor cells (IPCs), a type of transit amplifying cells. The Tbr2+ IPCs were highly responsive to neurogenic stimuli, more than doubling after voluntary wheel running. Notably, the Tbr2+ IPCs formed cellular clusters, the average size of which (Tbr2+ cells per cluster) likewise more than doubled in runners. Conversely, Tbr2+ IPCs were selectively depleted by antimitotic drugs, known to suppress neurogenesis. After cessation of antimitotic treatment, recovery of neurogenesis was paralleled by recovery of Tbr2+ IPCs, including a transient rebound above baseline numbers. Finally, Tbr2 was examined in the context of additional TFs that, together, define a TF cascade in embryonic neocortical neurogenesis (Pax6 → Ngn2 → Tbr2 → NeuroD → Tbr1). Remarkably, the same TF cascade was found to be linked to stages of neuronal lineage progression in adult SGZ. These results suggest that Tbr2+ IPCs play a major role in the regulation of adult hippocampal neurogenesis, and that a similar transcriptional program controls neurogenesis in adult SGZ as in embryonic cerebral cortex."

(Vik-Mo et al., 2012)

The Role of Neural Stem Cells in Neurorestoration

"In the dentate gyrus cells migrate only a few micrometers, from the subgranular zone to the granule cell layer. Cells develop from a precursor cell type in which mitotic events are found"

"In mammals the SVZ contains three cell populations important for stem cell proliferation. The proper stem cell population is maintained through slowly dividing astrocyte-like neural stem cells known as type B cells. These cells give rise to actively proliferating type C cells, which in turn give rise to immature neuroblasts, called type A cells. These neuroblasts, not yet neuronally committed, migrate to the olfactory bulb via chain migration by cell-cell contacts."

(Stoll, 2014)

Advances toward regenerative medicine in the central nervous system: challenges in making stem cell therapy a viable clinical strategy

"In the adult dentate gyrus, cells proliferate in the inner granule cell layer, and migrate radially outward as they differentiate (Zhao et al., 2006). New granule cells in hippocampal dentate gyrus receive primary glutamatergic afferents from entorhinal cortex and project axons to inhibitory interneurons and pyramidal cells of area CA3"

(Zhao et al., 2006)

Distinct Morphological Stages of Dentate Granule Neuron Maturation in the Adult Mouse Hippocampus

"Adult neurogenesis in the dentate gyrus may contribute to hippocampus-dependent functions, yet little is known about when and how newborn neurons are functional because of limited information about the time course of their connectivity. By using retrovirus-mediated gene transduction, we followed the dendritic and axonal growth of adult-born neurons in the mouse dentate gyrus and

identified distinct morphological stages that may indicate different levels of connectivity. Axonal projections of newborn neurons reach the CA3 area 10–11 d after viral infection, 5–6 d before the first spines are formed. Quantitative analyses show that the peak of spine growth occurs during the first 3–4 weeks, but further structural modifications of newborn neurons take place for months. Moreover, the morphological maturation is differentially affected by age and experience, as shown by comparisons between adult and postnatal brains and between housing conditions. Our study reveals the key morphological transitions of newborn granule neurons during their course of maturation."

Piriform Cortex Immature Neurons

The piriform cortex contains immature excitatory neurons in layer II that express DCX but are generated during prenatal development rather than through adult neurogenesis. These cells can be classified into distinct morphological subtypes including tangled cells and complex cells based on their degree of maturation. (6 sources)

The piriform cortex represents a unique case among brain regions containing immature neurons, as the DCX-expressing cells found in layer II are not products of adult neurogenesis but rather neurons generated during embryonic development that maintain immature characteristics for extended periods (Rotheneichner et al., 2018) (Bartkowska et al., 2022). These cells expressing the immature neuronal marker doublecortin (DCX) and the polysialylated-neural cell adhesion molecule reside in layer II of the piriform cortex and progressively resume their maturation into glutamatergic (TBR1+, CaMKII+) neurons (Rotheneichner et al., 2018).

The classical nomenclature for piriform cortex immature neurons includes two main morphological subtypes based on their degree of maturation (Coviello et al., 2021). Tangled cells represent the least mature subtype, characterized as very small cells with very short and intricate processes (Coviello et al., 2021) (Gomez-Climent et al., 2008). Complex cells are more mature and display axon initial segments (AIS) and typical apical dendrites extending into layer I (Coviello et al., 2021). Additional transitional forms include semilunar-pyramidal transitional neurons and scarce cells resembling semilunar, pyramidal, and fusiform neurons (Gomez-Climent et al., 2008).

These immature neurons are generated primarily during embryonic development, with most cells produced around embryonic day 15.5, and they express DCX for extended periods lasting 2-3 months after birth (Bartkowska et al., 2022) (Yang et al., 2015). Unlike the dentate gyrus where DCX expression is transient in newly generated neurons, piriform cortex DCX+ cells appear to be in a prolonged immature state and are associated with synaptic plasticity rather than continuous neurogenesis (Merz et al., 2013) (Gomez-Climent et al., 2008).

Evidence

(Rotheneichner et al., 2018)

Cellular Plasticity in the Adult Murine Piriform Cortex: Continuous Maturation of Dormant Precursors Into Excitatory Neurons

"cells expressing the immature neuronal marker doublecortin (DCX) and the polysialylated-neural cell adhesion molecule reside in layer II of the piriform cortex"

"these immature neurons located in the murine piriform cortex"

"progressively resume their maturation into glutamatergic (TBR1+, CaMKII+) neurons"

"immature neurons found in layer II of the piriform cortex are generated prenatally"

(Bartkowska et al., 2022)

Postnatal and Adult Neurogenesis in Mammals, Including Marsupials

"DCX-expressing cells in the piriform cortex of adult mice express DCX, indicating the presence of immature neurons"

"DCX-expressing cells are generated during embryonic development of the piriform cortex and express DCX for a long period (2-3 months after birth) (Nácher et al., 2001)(Yang et al., 2015). During embryonic development, DCX is expressed in progenitor cells and migrating neuronal-lineage cells, while in adult brains, transient expression of DCX is present in immature neurons"

"The majority of proliferated cells from the SVZ migrate to the olfactory bulb, whereas, in the dentate gyrus, cells reside within this

structure after division and differentiation into neurons"

"the generation of granule cells in the dentate gyrus, olfactory bulb, and cerebellum of eutherian species"

(Coviello et al., 2021)

PSA Depletion Induces the Differentiation of Immature Neurons in the Piriform Cortex of Adult Mice

"Immature neurons in the PCX layer II can be classified into different subtypes depending on their morphology, physiological properties and degree of maturation (Gómez-Climent et al., 2008), namely: (1) tangled cells, which are very small and have very short and intricate processes; (2) complex cells, which are more mature and display axon initial segments (AIS) and typical apical dendrites in layer I [15]."

(Gomez-Climent et al., 2008)

A population of prenatally generated cells in the rat paleocortex maintains an immature neuronal phenotype into adulthood.

"New neurons in the adult brain transiently express molecules related to neuronal development, such as the polysialylated form of neural cell adhesion molecule, or doublecortin (DCX). These molecules are also expressed by a cell population in the rat paleocortex layer II, whose origin, phenotype, and function are not clearly understood. We have classified most of these cells as a new cell type termed tangled cell. Some cells with the morphology of semilunar-pyramidal transitional neurons were also found among this population, as well as some scarce cells resembling semilunar, pyramidal. and fusiform neurons. We have found that none of these cells in layer II express markers of glial cells, mature, inhibitory, or principal neurons. They appear to be in a prolonged immature state, confirmed by the coexpression of DCX, TOAD/Ulip/CRMP-4, A3 subunit of the cyclic nucleotide-gated channel, or phosphorylated cyclic adenosine monophosphate response element-binding protein. Moreover, most of them lack synaptic contacts, are covered by astroglial lamellae, and fail to express cellular activity markers, such as c-Fos or Arc, and N-methyl-d-aspartate or glucocorticoid receptors. We have found that none of these cells appear to be generated during adulthood or early youth and that most of them have been generated during embryonic development, mainly in E15.5."

(Yang et al., 2015)

Prenatal genesis of layer II doublecortin expressing neurons in neonatal and young adult guinea pig cerebral cortex

"Cells expressing doublecortin (DCX+) occur at cortical layer II, predominantly over the paleocortex in mice/rats, but also across the neocortex among larger mammals. Here, we explored the time of origin of these cells in neonatal and 2-month-old guinea pigs following prenatal BrdU pulse-chasing. In the neocortex, BrdU+ cells birth-dated at embryonic day 21 (E21), E28, and E35 laminated over the cortical plate with an inside-out order. In the piriform cortex, cells generated at E21 and E28 occurred with a greater density in layer II than III. Many cells were generated at later time points until birth, occurring in the cortex without a laminar preference. DCX+ cells in the neocortex and piriform cortex partially co-colocalized with BrdU (up to 7.5%) in the newborns after pulse-chasing from E21 to E49 and in the 2 month-old animals after pulse-chasing from E28 to E60/61, with higher rates seen among the E21-E35 groups. Together, layer II DCX+ cells in neonatal and young adult guinea pigs may be produced over a wide prenatal time window, but mainly during the early phases of corticogenesis. Our data also show an earlier establishment of the basic lamination in the piriform relative to neocortical areas in guinea pigs."

(Merz et al., 2013)

Evidence that Doublecortin Is Dispensable for the Development of Adult Born Neurons in Mice

"DCX is almost exclusively expressed in immature neurons of the two neurogenic niches, the hippocampal dentate gyrus (DG) and the subventricular zone (SVZ; Figure 1). Some postmitotic cells of the piriform cortex express DCX. A subpopulation of these cells retains immature characteristics and is associated with synaptic plasticity (Klempin et al., 0)"

"adult neural stem cells give rise to new hippocampal dentate granule neurons and interneurons of the olfactory bulb throughout life"

"migrating neuroblasts and immature neurons"

"newly generated dentate granule neurons"

Olfactory Bulb Immature Neurons

The olfactory bulb receives immature inhibitory neurons called neuroblasts that migrate from the subventricular zone through the rostral migratory stream. These neuroblasts differentiate into two main types of GABAergic interneurons: granule cells and periglomerular neurons. (7 sources)

The olfactory bulb represents the primary destination for adult-generated inhibitory interneurons, which originate from the subventricular zone (SVZ) neurogenic niche (Delgado-Garcia, 2016) (Doetsch et al., 1997). The SVZ contains three main cell populations: slowly dividing astrocyte-like neural stem cells (type B cells),

actively proliferating transit amplifying cells (type C cells), and immature neuroblasts (type A cells) (Vik-Mo et al., 2012) (Delgado-Garcia, 2016). Type B cells give rise to type C cells, which in turn generate the migratory type A neuroblasts that are not yet neuronally committed (Vik-Mo et al., 2012).

The migration process involves a highly organized journey through the rostral migratory stream (RMS), where type A neuroblasts form chains and migrate toward the olfactory bulb through a tube formed by astrocytes (Regalado-Santiago et al., 2016) (Hussain et al., 2023). Upon reaching the olfactory bulb, these immature neurons leave the RMS and migrate radially toward the glomeruli, where they differentiate into different subtypes of interneurons (Regalado-Santiago et al., 2016).

The classical nomenclature identifies two main types of interneurons that develop from these migrating neuroblasts. The majority differentiate into GABAergic granule neurons located in the granule cell layer, while others become GABAergic periglomerular neurons in the glomerular layer (Velusamy et al., 2017) (Hussain et al., 2023) (Delgado-Garcia, 2016). These newly integrated interneurons provide lateral inhibition: periglomerular neurons inhibit between individual glomeruli, while granule cells provide lateral inhibition between mitral and tufted cells (Delgado-Garcia, 2016). The developmental timeline shows that synaptic integration occurs around 21 days after neuroblast generation, with younger immature neurons at 10 days not yet synaptically integrated (Vangeneugden et al., 2015).

Evidence

(Delgado-Garcia, 2016)

Adult Brain Neurogenesis, Neural Stem Cells and Neurogenic Niches

> *"The SVZ is the layer along the lateral ventricle. There are three main cell types in the SVZ neurogenic niche: neuroblasts (Type A cells), NSC-SVZ astrocytes (Type B cells) and rapidly dividing transit amplifying immature progenitors (Type C cells). Type B and C cells are precursor cells, of those, Type B cells are the primary precursors or NSCs that divide to give rise to Type C cells or rapidly dividing transit amplifying cells or proliferating progenitors, which in turn generate migratory Type A cells or neuroblasts [52][53][54][55][56][57]. Neuroblasts of the SVZ join the network in the rostral migratory stream (RMS) to reach the olfactory bulb, where differentiate into granular and glomerular neurons [52][53][54][55][56][57]. In the olfactory bulb, primary sensory neurons project to glomeruli where they synapse onto mitral and tufted cells, which in turn relay information to the olfactory cortex, periglomerular neurons provide lateral inhibition between individual glomeruli, whereas granule cells provide lateral inhibition between mitral and tufted cells [47]. The SGZ is a germinal layer between the dentate gyrus and the hilus in the hippocampus. NSCs generate D1 cells, which divide to generate D2 cells or neuroblasts. D2 cells reach the granule layer becoming D3 cells, which mature into new granule neurons [57][58][59]."*

(Doetsch et al., 1997)

Cellular Composition and Three-Dimensional Organization of the Subventricular Germinal Zone in the Adult Mammalian Brain

> *"The adult mammalian subventricular zone (SVZ) contains stem cells that give rise to neurons and glia. In vivo, SVZ progeny migrate 3–8 mm to the olfactory bulb, where they form neurons. We show here that the SVZ of the lateral wall of the lateral ventricles in adult mice is composed of neuroblasts, glial cells, and a novel putative precursor cell. The topographical organization of these cells suggests how neurogenesis and migration are integrated in this region. Type A cells had the ultrastructure of migrating neuronal precursors. These cells were arranged as chains parallel to the walls of the ventricle and were polysialylated neural adhesion cell molecule- (PSA–NCAM), TuJ1- (β-tubulin), and nestin-positive but GFAP- and vimentin-negative. Chains of Type A cells were ensheathed by two ultrastructurally distinct astrocytes (Type B1 and B2) that were GFAP-, vimentin-, and nestin-positive but PSA–NCAM- and TuJ1-negative. Type A and B2 (but not B1) cells incorporated [3H]thymidine. The most actively dividing cell in the SVZ corresponded to Type C cells, which had immature ultrastructural characteristics and were nestin-positive but negative to the other markers. Type C cells formed focal clusters closely associated with chains of Type A cells. Whereas Type C cells were present throughout the SVZ, they were not found in the rostral migratory stream that links the SVZ with the olfactory bulb. These results suggest that chains of migrating neuroblasts in the SVZ may be derived from Type C cells. Our results provide a topographical model for the adult SVZ and should serve as a basis for the in vivo identification of stem cells in the adult mammalian brain."*

(Vik-Mo et al., 2012)

The Role of Neural Stem Cells in Neurorestoration

> *"In the dentate gyrus cells migrate only a few micrometers, from the subgranular zone to the granule cell layer. Cells develop from a precursor cell type in which mitotic events are found"*
>
> *"In mammals the SVZ contains three cell populations important for stem cell proliferation. The proper stem cell population is*

maintained through slowly dividing astrocyte-like neural stem cells known as type B cells. These cells give rise to actively proliferating type C cells, which in turn give rise to immature neuroblasts, called type A cells. These neuroblasts, not yet neuronally committed, migrate to the olfactory bulb via chain migration by cell-cell contacts."

(Regalado-Santiago et al., 2016)

Mimicking Neural Stem Cell Niche by Biocompatible Substrates

"In the rostral migratory stream (RMS), type A cells form a chain and migrate toward the olfactory bulb through a tube formed by astrocytes. Upon reaching the olfactory bulb, immature neurons leave the RMS and migrate radially toward the glomeruli, where they differentiate into different subtypes of interneurons that, finally, are synaptically integrated"

"hippocampal neurogenesis in the DG begins, as in SVZ neurogenesis, with the proliferation of radial (type I cell) and nonradial (type II cell) precursors that give rise to intermediate progenitors, which in turn generate neuroblasts"

"The new immature neurons move into the inner granule cell layer and differentiate into dentate granule cells in the hippocampus."

(Hussain et al., 2023)

Adult neurogenesis: a real hope or a delusion?

"In the rostral migratory stream, neuroblasts continue to chain together before migrating toward the olfactory bulb through a tube created by astrocytes. These immature neurons grow into intermediate neurons near the center of the olfactory bulb. They primarily develop into GABAergic granule neurons, while some also become GABAergic periglomerular neurons."

(Velusamy et al., 2017)

Protective Effect of Antioxidants on Neuronal Dysfunction and Plasticity in Huntington's Disease

"In the hippocampal SGZ, proliferating NSCs develop into intermediate progenitors, which generate neuroblasts or immature neurons. These newly generated immature neurons migrate into the inner granule cell layer (GCL) and differentiate into new granule neurons of the hippocampus (Ming et al., 2005)"

"In the SVZ, proliferating radial glia-like cells give rise to transient amplifying cells that generate migrating neuroblasts (Ming et al., 2005)(Lois et al., 1996). Through the rostral migratory system (RMS), neuroblasts migrate towards the olfactory bulb (OB) (Lois et al., 1996). In the OB, immature neurons differentiate into different subtypes of interneurons in granule cell layer (GCL) and glomerular layer (GLOM)."

(Vangeneugden et al., 2015)

Commentary: “Posttraining ablation of adult-generated olfactory granule cells degrades odor-reward memories”

"new immature neurons are continuously produced and then migrate out to their respective target circuits, differentiating either into glutamatergic neurons (dentate gyrus) or into mostly GABAergic interneurons called granule cells in the olfactory bulb"

"the tagged population 21 days following TAM injection included both 21 days-old OGC as well as younger immature neurons born during these last 21 days"

"synaptically integrated 21 days-old, but not immature 10 days-old neurons"

Striatum Immature Neurons

The striatum receives immature inhibitory neurons that migrate from the subventricular zone, as evidenced by the presence of BrdU and doublecortin-positive cells. However, compared to other brain regions, the striatum appears to have limited adult neurogenesis with relatively few immature neurons successfully integrating. (1 source)

The striatum represents another target region for immature neurons originating from the subventricular zone neurogenic niche. Evidence for striatal neurogenesis comes from the detection of BrdU and doublecortin-immunoreactive (BrdU+/DCX+) cells within the striatum, suggesting active migration of immature neurons from the subventricular zone (Dayer et al., 2005). These immature neurons are presumed to be inhibitory interneurons, similar to those that migrate to the olfactory bulb, as they originate from the same SVZ neurogenic source that produces GABAergic cell types (Model-Generated).

The classical nomenclature for striatal immature neurons is less well-established compared to other neurogenic regions, partly because adult striatal neurogenesis remains more controversial and appears to be quantitatively limited compared to the robust neurogenesis observed in the dentate gyrus and olfactory bulb (Model-Generated). The DCX-expressing cells found in the striatum likely represent neuroblasts at various

stages of migration and early differentiation, though their final interneuron subtypes and developmental progression require further characterization (Model-Generated).

Evidence

(Dayer et al., 2005)

New GABAergic interneurons in the adult neocortex and striatum are generated from different precursors

> *"BrdU and doublecortin-immunoreactive (BrdU+/DCX+) cells were seen within the striatum, suggesting migration of immature neurons from the subventricular zone. Surprisingly, no DCX+ cells were found within the neocortex."*

Cerebral Cortex Immature Neurons

The cerebral cortex shows limited evidence for immature neurons in most mammals, with some studies finding no DCX+ cells in neocortex while others report DCX-labeled immature neurons specifically in layer II of limbic cortices. The presence and significance of cortical immature neurons remains controversial and species-dependent. (2 sources)

The cerebral cortex presents a complex and controversial picture regarding the presence of immature neurons, with conflicting evidence across different studies and species. In many mammalian studies, no DCX+ cells were found within the neocortex, suggesting that unlike other brain regions, the cerebral cortex does not receive significant numbers of immature neurons from neurogenic niches (Dayer et al., 2005). This absence contrasts sharply with the robust populations of immature neurons found in the dentate gyrus, olfactory bulb, and other neurogenic regions.

However, some evidence suggests that certain cortical areas, particularly limbic cortices, may contain immature neurons under specific conditions or in certain species. In degus, significant numbers of DCX expressing neurons have been found in areas outside of the traditional neurogenic zones, with many areas of neocortex demonstrating DCX-labeled neurons in layer II, particularly in limbic cortices (Groen et al., 2021). These DCX-labeled cells do not stain with NeuN, confirming they are immature neurons (Groen et al., 2021).

The classical nomenclature for cortical immature neurons remains poorly defined due to the limited and controversial nature of cortical neurogenesis in most mammalian species. When present, these cells appear to be predominantly located in layer II and are characterized by their DCX expression and lack of mature neuronal markers like NeuN. The developmental origin, migration patterns, and final cell fates of these potential cortical immature neurons require further investigation to establish their biological significance and proper classification within the existing nomenclature systems.

Evidence

(Dayer et al., 2005)

New GABAergic interneurons in the adult neocortex and striatum are generated from different precursors

> *"BrdU and doublecortin-immunoreactive (BrdU+/DCX+) cells were seen within the striatum, suggesting migration of immature neurons from the subventricular zone. Surprisingly, no DCX+ cells were found within the neocortex."*

(Groen et al., 2021)

Widespread Doublecortin Expression in the Cerebral Cortex of the Octodon degus

> *"newly born neurons in the subgranular layer (SGL) of the dentate gyrus (DG), and in the subventricular zone (SVZ). The neurons generated in the SVZ migrate through the rostral migratory stream (RMS) to the olfactory bulb"*

> *"DCX-labeled neurons are predominantly present in the SGL and SVZ, with only a few DCX neurons present in piriform cortex"*

> *"significant numbers of DCX expressing neurons in areas outside of SVZ, DG, and PC. Many areas of neocortex in degu demonstrate DCX-labeled neurons in layer II, and most of these neurons are found in the limbic cortices. The DCX-labeled cells do not stain with NeuN, indicating they are immature neurons."*